Using machine learning to assess the livelihood impact of electricity access

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Nathan Ratledge[1,2], Gabe Cadamuro[3], Brandon de la Cuesta[4], Matthieu Stigler[5] & Marshall Burke[6,7,8] ✉

In many regions of the world, sparse data on key economic outcomes inhibit the development, targeting and evaluation of public policy[1,2]. We demonstrate how advancements in satellite imagery and machine learning (ML) can help ameliorate these data and inference challenges. In the context of an expansion of the electrical grid across Uganda, we show how a combination of satellite imagery and computer vision can be used to develop local-level livelihood measurements appropriate for inferring the causal impact of electricity access on livelihoods. We then show how ML-based inference techniques deliver more reliable estimates of the causal impact of electrification than traditional alternatives when applied to these data. We estimate that grid access improves village-level asset wealth in rural Uganda by up to 0.15 standard deviations, more than doubling the growth rate during our study period relative to untreated areas. Our results provide country-scale evidence on the impact of grid-based infrastructure investment and our methods provide a low-cost, generalizable approach to future policy evaluation in data-sparse environments.

Accurate public policy evaluation requires data on outcomes of interest that could be shaped by a given policy and an ability to isolate the impact of the policy from other correlated changes that might also affect these outcomes. Sparse data on key livelihood outcomes across much of the world continues to hinder such evaluations[2]. In many countries, existing data provide only an intermittent or aggregate snapshot of well-being, and the dearth of granular and reliable livelihood data over time makes it challenging to both measure changes in well-being and correctly attribute them to policy interventions. Left unaddressed, this data shortage will continue to impede understanding of what interventions best improve livelihoods in some of the most impoverished regions of the world.

The combination of recent advancements in computer vision and a growing abundance of satellite imagery have recently shown promise in improving the measurement of livelihoods in data-sparse environments[2–6]. Deep-learning models trained on daytime satellite imagery and nighttime lights are now capable of predicting up to 75% of the variation in local-level economic data on which the model was not trained[3,6,7]. However, with few exceptions (such as ref. [8]), these new approaches have yet to be deployed in the evaluation of policies or interventions designed to improve livelihoods[2].

Here we develop a generalizable method that overcomes several data and inference challenges and apply it to estimate the causal impact of electrification on livelihoods in Uganda. With a population of 45 million, Uganda has had recent success extending electricity access—from 12% coverage in 2010 to 41% in 2019 (ref. [9])—yet little is known about the economic impact of this country-wide investment. Similar conditions exist in many lower-income regions across Africa and the Global South, and the resulting challenges are also typical of other regions. Nearly 1 billion people globally and 600 million in sub-Saharan Africa (SSA) lack access to modern electricity[10,11] (Fig. 1a), and the approximately $20 billion invested annually in expanding electricity access in SSA in recent years is only a fraction of the estimated $45–100+ billion needed annually over the next 20 years to secure minimum levels of continent-wide access[11]. Such investments are suggested to generate several potential benefits, including improved household-level productivity through increased lighting and use of electrical appliances and improved community-level productivity through new business development and extended hours[12,13]. Electricity could also improve the functioning of public health facilities, reduce indoor air pollution by substituting for dirtier energy sources used for cooking, heating or lighting and improve educational outcomes by enhancing at-home or in-school learning.

However, empirical evidence on the impact of grid expansions in lower-income regions remains limited, particularly in SSA. Although an early paper found positive effects of grid electrification on employment and earnings[14], more recent experimental work found limited impacts on traditional poverty indicators[12,15]. Other studies note the continued lack of evidence for policymakers and call for more research[16–18]. Understanding where and how expanded electricity access is beneficial will be critical for guiding the proposed $1–2.5+ trillion investment in SSA electrification efforts in the coming decades[11] and meeting the United Nations' goal of global energy access.

Successfully deploying remote sensing and ML to measure policy impact involves surmounting at least four challenges. First, as most policy interventions are not randomized, reliable causal effect estimates typically require observation of both treated and untreated units (for example, individuals or locations) before and after the policy was instituted. Generating such longitudinal data from a remote-sensing and deep-learning pipeline, particularly at a scale and resolution

[1]Emmett Interdisciplinary Program in Environment and Resources, Stanford University, Palo Alto, CA, USA. [2]TomKat Center for Sustainable Energy, Stanford University, Palo Alto, CA, USA. [3]Atlas AI, Palo Alto, CA, USA. [4]King Center on Global Development, Stanford University, Palo Alto, CA, USA. [5]Environmental Policy Lab, ETH Zurich, Zurich, Switzerland. [6]Center on Food Security and the Environment, Stanford University, Palo Alto, CA, USA. [7]Doerr School of Sustainability, Stanford University, Palo Alto, CA, USA. [8]National Bureau of Economic Research, Cambridge, MA, USA. ✉e-mail: mburke@stanford.edu

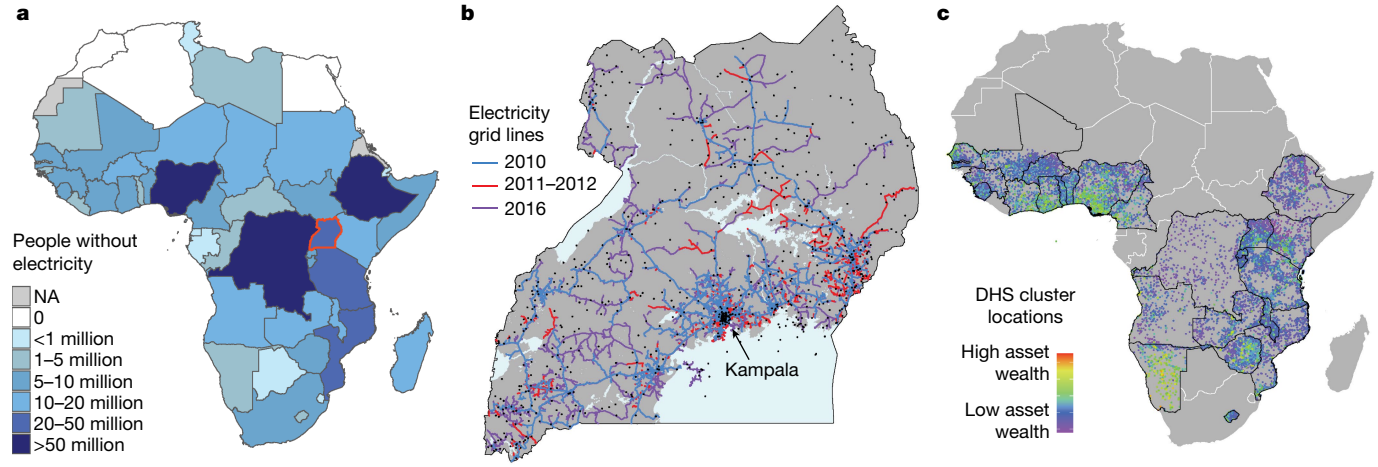

Fig. 1 | Electricity access rates, Ugandan electrification and asset wealth training data. a, Nearly 600 million people in SSA lacked access to central-grid-based electricity in 2018. Countries are coloured by the number of people lacking access to electricity. Uganda, with roughly 24 million people without grid power in 2018, is highlighted by a red border. **b**, The electricity grid has expanded rapidly in Uganda in recent years, including to previously unelectrified regions of the country. Black dots illustrate the 641 EAs from the 2016 DHS. **c**, Wealth indexes for 27,174 survey EAs (roughly, villages) used in model training are coloured from low asset wealth to high asset wealth. Data cover 25 countries over 13 years and represent 641,621 household surveys.

necessary to evaluate the causal effect of a policy, is challenging and has not been the focus of previous work[2]. A second concern is that image-derived proxies for a dependent variable of interest could themselves include the policy intervention of interest, undermining inference. For example, poverty predictions from a satellite-based model cannot be reliably used to study the impact of new road construction on poverty if there is a chance that the model looks for a road to decide whether a location is poor. Third, for continuous outcomes, the distributions of predicted outcomes from both simple linear models and more complex deep-learning models commonly have lower variance than the distribution of observed outcomes. Models seeking to predict economic well-being, for example, often overpredict for poorer individuals and underpredict for wealthier ones; this can bias estimates of treatment effects, as we show. Similarly, recipients of interventions often have pre-treatment outcomes that are evolving differentially relative to untargeted populations—for instance, road-improvement projects could be targeted to areas with rapid economic growth. As a result, benchmark research designs, such as difference-in-differences (DiD), can struggle to recover valid treatment effect estimates, even when outcomes are reliably measured[19,20].

To overcome these challenges in our Ugandan context, we created georeferenced data on the multi-year expansion of the electricity distribution grid throughout the country by digitizing historic maps for the years 2005–2018 and then combined them with satellite-based predictions of local-level asset wealth from a ML model trained on data from more than 27,000 villages across SSA (Fig. 1b,c and Methods; research pipeline depicted in Extended Data Fig. 1). We focus on asset wealth rather than other livelihood measurements (such as consumption expenditure), as asset wealth is considered a less noisy measure of households' longer-run economic well-being, is a common component of multi-dimensional livelihood measures used around the world[21,22] and is consistently measured in many georeferenced, nationally representative surveys conducted in SSA. Unfortunately, such surveys often do not revisit the same households or locations across survey waves, making it difficult to construct the repeated local-level measurements on which many causal inference techniques depend.

Following recent approaches[3,5], we assemble survey-based household asset wealth data from Demographic and Health Surveys (DHS) of 641,621 households in 27,174 enumeration areas (EAs or 'cluster', analogous to village or neighbourhood) across 25 countries in SSA during a period of 13 years (Fig. 1c and Extended Data Fig. 2) and use principal component analysis to construct an asset wealth index from survey responses of households to questions about ownership of specific assets (Methods and Extended Data Fig. 3), using data from all households in the sample to construct one common index. Wealth index estimates are robust to alternative approaches to index construction (Extended Data Fig. 4). We then average wealth index estimates within each EA, the level at which geocoordinates are available in the survey data.

We then train a convolutional neural network (CNN) to predict wealth index values at the cluster level from temporally and spatially matched multispectral satellite images. We use a ResNet-18 architecture[23] with the input layer modified to include further satellite bands to supplement RGB imagery and the final layer modified to provide a single scalar estimate of the wealth composite. We split data into disjoint training (60%), validation (20%) and test (20%) sets, with the model evaluated on the held-out test data (Methods). We evaluate CNN models trained with and without data from Uganda in the training and validation sets; the latter setting replicates a common real-world situation in which ground-truth data from a target geography of interest may not exist.

To overcome the concern that visual indicators of the independent variable of interest (electricity distribution grid) could be used by the CNN in its prediction of the wealth index and thus generate a mechanical relationship between grid expansion and asset wealth, we departed from earlier work and excluded measures of self-reported household electrification from the construction of the wealth index (Extended Data Fig. 4) and used only daytime satellite imagery rather than nighttime lights imagery as input to the CNN; nighttime imagery can directly pick up the presence of local electrification and the coarse (30-metre) daytime imagery that we use is unlikely to directly detect electrification infrastructure.

In our setting, the use of standard mean squared error (MSE) loss functions in CNNs (or in simpler linear models) generates a distribution of model-predicted values that is lower in variance than the true distribution (Extended Data Fig. 5), with the model overpredicting wealth at the lower end of the wealth distribution and underpredicting in wealthier regions. We show mathematically and through simulation that this bias is consequential for downstream inference tasks, attenuating (biasing towards zero) treatment effect estimates that use these predictions to measure outcomes (Methods and Extended Data Fig. 6). To mitigate these biases, we modify the standard MSE loss function with an extra term that penalizes bias in each quintile of the wealth distribution (Methods). This loss function generates wealth predictions that

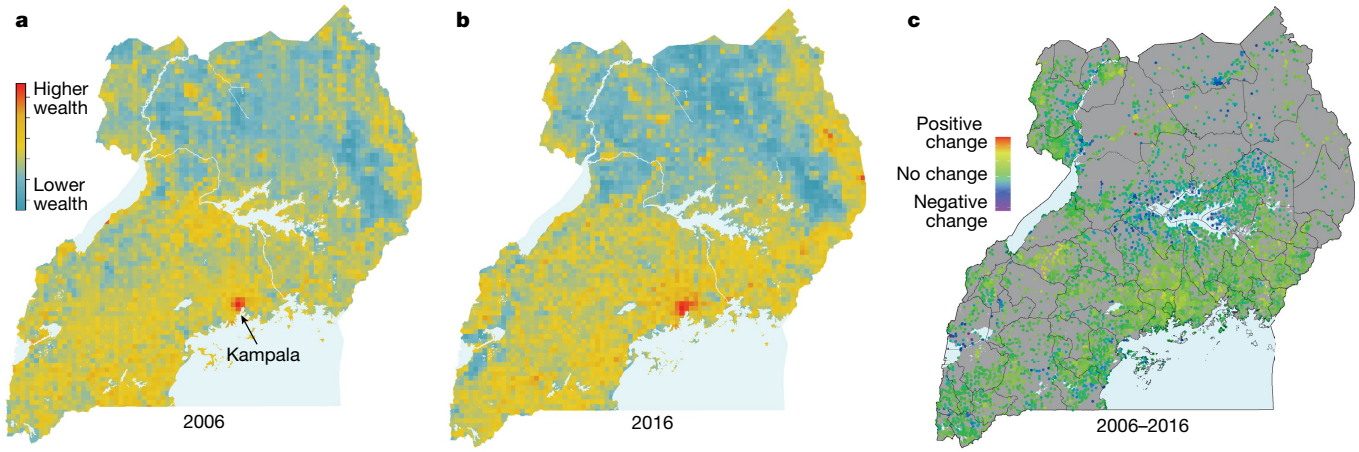

Fig. 2 | Wealth predictions in Uganda over time. a,**b**, Asset-based wealth index values predicted with a CNN across Uganda in 2006 and 2016, the first and last years, respectively, of our analysis. **c**, Change in asset wealth between 2006 and 2016 for 6,900 populated villages and neighbourhoods. Asset-based wealth increased by an average of 0.15 standard deviations during the period across observed locations.

substantially reduce bias in downstream inference tasks (Extended Data Fig. 5) but at the cost of slightly lower predictive performance as measured by r^2, a common test statistic of interest in this setting[2,3]. This result highlights that the common practice of maximizing average predictive performance across the output distribution (for example, for a regression problem, maximizing the r^2 between predicted and observed values in a test set) can actually worsen performance on key downstream tasks.

Our custom loss function helps ensure that bias in outcome prediction will not lead to bias in estimated treatment effects. But accurate outcome data alone do not solve the separate, more conventional inference problem in which a given unit is only observed in either a treated or an untreated state, and the unobserved counterfactual state must be inferred to estimate the effect of treatment. To solve this inference task, we exploit two recent ML-based causal inference approaches, matrix completion (MC) and synthetic controls with elastic net (SC-EN), which have been shown to be more robust in the face of common threats to inference, such as non-parallel trends in pre-treatment outcomes[24,25]. MC works by treating counterfactual untreated observations in the treatment group as missing values in a matrix, with these values imputed through a regularized process that penalizes matrix complexity[24] (Methods). SC-EN uses a flexible approach to matching treated units with a weighted average of control units that were trending similarly pre-treatment[25]. We use cross-validation to show that MC and SC-EN estimators are capable of predicting held-out observations in control locations (Methods and Extended Data Fig. 6) and use simulation to show that ML-based causal estimates are less biased than DiD estimators in the presence of time-trending unobservables (Methods and Extended Data Fig. 7). To address a final potential threat to inference—the possibility that other infrastructure investments were rolled out simultaneously with investments in electrification—we compile and harmonize available data on road construction and on cellular phone ownership and coverage[26–28], key infrastructure investments that might also improve livelihoods, to test whether expanded electrification access was spatially and temporally coincident with these other public or private investments (Methods).

We then estimate causal impacts in three separate scenarios. In our base case ('with Uganda'), DHS data from Uganda contribute to the training and validation data used to train the CNN, mimicking a setting in which some local data are available to train models. In our 'without Uganda' case, we train on DHS data from African countries other than Uganda, mimicking a setting in which no local data are available for model development in the target geography. Predictions in these locations are made for all study years, 2006–2016. In the third ('full country') case, instead of filling in missing years for the available DHS locations, we use the CNN to predict wealth at 1 km over all Uganda and all study years (Fig. 2).

To estimate livelihood impacts of electrification expansion, we focus on locations that received grid access in 2011 or 2012, a period of rapid grid expansion during which we can observe outcomes for many years both before and after expansion. We define 'treated' communities as those that were within 2 km of new distribution lines in 2011 or 2012. Control communities are defined as those that did not receive grid access by 2016, the most recent year for which DHS data are available (see Methods). Of about 6,900 community locations, 3,601 are unelectrified by 2016 and are used as controls, and 268 receive electrification during the 2011–2012 period and are used as treated units, resulting in a total of 3,869 locations observed in the full-country analysis.

CNN predictive performance

The performance of our CNN model in predicting wealth across SSA consistently achieved r^2 values above 0.6 with a max of 0.68, thus explaining roughly two-thirds of the variation in held-out village-level DHS wealth measurements. Performance on this wealth prediction task is comparable with recent literature, which reports r^2 values between 0.6 and 0.75 and typically uses extra inputs (for example, nighttime lights) that we did not use[2,3]. Our preferred bias-corrected estimates achieved $r^2 = 0.64$ over all of SSA and 0.63 for Uganda clusters during model training. In the without-Uganda setting, we found slightly lower r^2 values when predicting on held-out Ugandan data, with an average r^2 of 0.50 within Uganda.

Full-country predictions of wealth levels and changes (Fig. 2) indicate that the north and east of the country remain the poorest parts, consistent with World Bank analyses[29]. They also indicate substantial wealth growth in central Uganda, in particular the peri-urban areas near the capital city of Kampala, and slower growth in northern regions with a recent legacy of conflict, consistent with recent studies[30]. On average, our village wealth indices increased by 0.19 units between 2006 and 2016, equal to a 0.18 standard deviation increase relative to the pooled Ugandan wealth distribution between 2006 and 2016.

Estimated impacts of electrification

MC and SC-EN estimate similar, statistically significant effects of electrification on wealth across our three data imputation settings

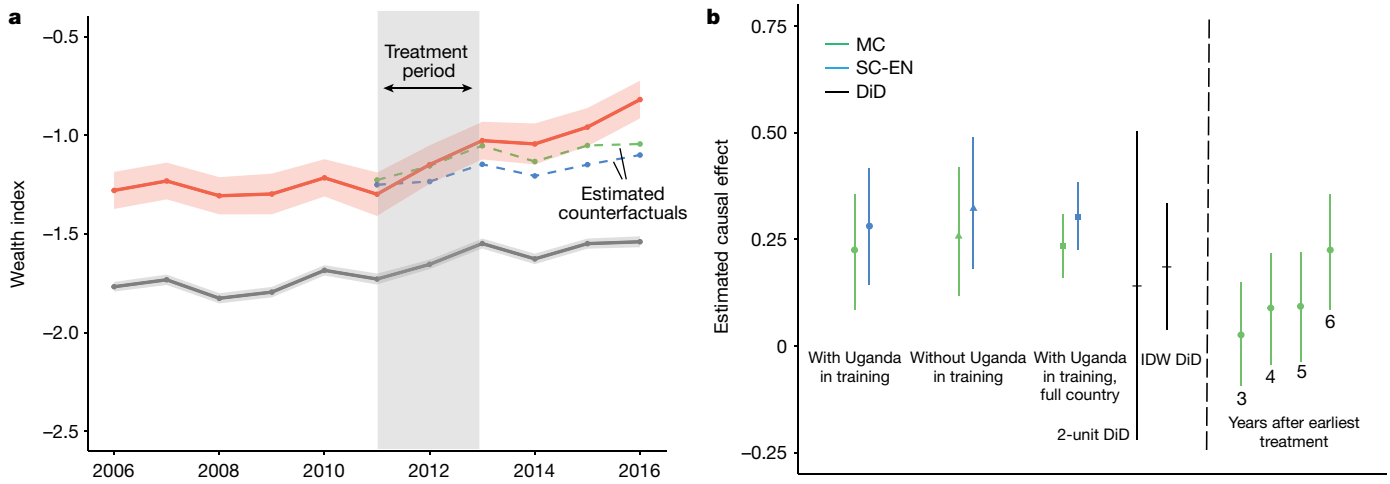

Fig. 3 | Electricity access increases household wealth in electrified communities compared with unelectrified communities. a, Solid lines show average asset wealth in untreated (grey) and treated (red) locations and shaded bands represent the standard error of the mean. Untreated counterfactuals for the treated group, as predicted by MC (green) and SC-EN (blue), are shown by dotted lines. **b**, Estimated causal effect of electrification on wealth by the end of the sample (2016). Error bars represent 95% CIs. CIs for each of the ML estimates are based on 500 bootstrapped model runs. We find similar statistically significant positive impacts in each of our three ML-based estimates (with Uganda, without Uganda and full country). '2-unit' represents a repeated cross-section DiD run using only DHS estimates. 'IDW' reflects DiD results from an inverse distance weighting approach. The four numbered lines on the right represent the causal estimates in the third to sixth years (2013–2016) after the initial treatment year, with CIs based on 100 bootstraps.

(Fig. 3). In the with-Uganda setting, the MC algorithm estimates a 0.23 (95% confidence interval (CI) 0.09, 0.36) unit improvement in asset-based wealth by 2016 for treated communities, equal to a 0.15 standard deviation increase relative to the sample wealth distribution. The results are robust to alternative inclusion buffers (Extended Data Fig. 8). As expected, point estimates are attenuated when the CNN prediction bias at different parts of the wealth distribution is less heavily penalized, producing a larger Berkson-type error (Extended Data Fig. 9). Treatment effects seem to increase over time (Fig. 3b), consistent with a commonly held hypothesis that it may take time for the impact of electrification to be realized[14,31].

We find no evidence that the rollout of the electrification grid was correlated with other key infrastructure investments—namely, road construction or cellular service—that could have independently affected local livelihoods. Available data show no change in Ugandan road coverage between 2007 (pre-treatment) and 2012 (post-treatment)[26], and recorded road-improvement projects were limited in scope and uncorrelated with electrification (Methods and Extended Data Fig. 10). Similarly, we find no relationship during our study period between grid expansion and household cellular phone ownership, local cellular phone coverage or coverage quality[28] (Extended Data Fig. 11), reinforcing the conclusion that other public and private investment projects were not colocated or timed with grid development.

Estimates from a model that did not use Uganda data in training are nearly identical to the with-Uganda case. Using wealth predictions across all of Uganda (n = 3,869 locations) yields a similar but much more precisely estimated effect (0.23, 95% CI 0.16, 0.31). The narrower CI in this latter setting reflects the larger amount of data available when all populated locations and years are used to estimate treatment effects. Both results indicate that our methodology can accurately estimate treatment effects even in the absence of training or validation data in the target geography.

We compare our main results with two DiD estimates that could have been implemented in the absence of our satellite-derived data: a 'two-unit' DiD in which outcomes for each treatment group and period are constructed as the simple average of wealth values across any treated or control unit observed in that period's survey (recall that the same locations are not observed across survey rounds) or an 'inverse-distance-weighted' (IDW) DiD that imputes wealth values in missing location-years as the IDW average of nearby treated or control surveyed locations in that year (Methods and Fig. 3b). We conduct a standard pre-treatment parallel-trends test and find violation of parallel pre-trends more often than would be expected by chance, meaning that DiD estimates could be biased (Extended Data Fig. 12). In the two-unit case, the treatment effect point estimate is lower than our main estimate and with a wide CI that contains zero. In the IDW case, the treatment effect estimate is similar in both magnitude and statistical significance to our main estimate (Fig. 3b).

We also compare our main results to output from a CNN model with a typical MSE loss function (that is, without the extra bias penalty); recall that this prediction model had a higher overall prediction r^2. Using predictions from this model, we estimate a 0.13 (95% CI 0.05, 0.24) effect of electrification for the with-Uganda case (Extended Data Fig. 9), or roughly 50% smaller than our preferred estimate. This result again highlights the importance of optimizing the prediction model for the downstream task.

Finally, to further understand how electrification affected ownership of specific assets contained in our asset index, we estimated the effect of electrification on ownership of specific assets using DHS data and two-period DiD (Methods and Extended Data Fig. 13). We find positive impacts across 8 of 11 variables, with the strongest being those related to home construction and appliances.

Discussion

Across various approaches to generating wealth estimates (with Uganda, without Uganda and full country) and two ML-based estimators (MC and SC-EN), our results provide evidence that villages that gain electricity grid access enjoy increases in asset wealth as compared with non-electrified communities. Our with-Uganda estimates suggest that treated communities grew more than twice as fast during the post-treatment period compared with unconnected communities. Our broader, full-country data setting suggests similar and more precisely estimated impacts on asset wealth. These accelerated growth rates for communities with new access to electricity are particularly notable in the communities that we study, which—at baseline—are some of the least wealthy on the African continent. Our results indicate that access to electrification can generate meaningful improvements in economic

well-being and adds evidence to the continuing but poorly informed[16] public policy debates concerning the costs and benefits of large-scale infrastructure development in lower-income regions.

Our findings contrast with some recent experimental work in SSA that found limited impact of household-level electrification on livelihoods[12,32]. One possible explanation for this difference is that we study, in effect, the household impact of community-level electrification, whereas the other work examined the impact of individual households gaining grid access. Community-level electrification could unlock a range of economic benefits and positive local labour-market spillovers that might not be fully observed in a household-level intervention[15]. A second explanation is that we are able to assess the evolution of the impact of electrification for substantially longer after treatment, up to 5 years (Fig. 3b), allowing for the potential to observe longer-term effects, such as new business creation and associated labour-market opportunities. This difference reinforces a distinct strength of our methodology, namely that estimates can be consistently revised with relative ease as new satellite imagery and grid data become available, and dynamic treatment effects can be estimated on large scales. Finally, data on household-level asset ownership suggest an uptake in products that require electricity after electrification, consistent with at least some homes gaining new access (Extended Data Fig. 13). With some exceptions[31], our results are more in keeping with the limited quasi-experimental work that found positive effects of community-level grid expansion, including in South Africa[14] and the Philippines[13].

Our results do not provide explicit insight on the many potential mechanisms through which expansion of the electricity grid could improve household asset wealth, including increased household productivity, new business opportunities or improved health. Insights on mechanism could potentially come by combining our comprehensive outcome data with information on key mediating variables (for example, labour force participation, agricultural incomes etc.), although the assumptions necessary for formal mediation analysis are probably strong. We also note that reliable electricity access could bring other important livelihood benefits that our results do not quantify, including improved safety and access to information and entertainment.

More broadly, our findings illustrate that careful combination of satellite imagery and deep learning can generate data appropriate for use in downstream causal inference tasks, including programme evaluation. We believe that our two-stage ML strategy—combining imagery and deep learning to generate outcome measurements and applying new ML-based causal inference estimators to these data to estimate programme treatment effects—represents a scalable, generalizable, repeatable and relatively low cost strategy for expanding policy evaluation in data-sparse environments, especially those in which such evaluation was previously cost-prohibitive or infeasible. We show that our strategy performs well even in a challenging setting in which no data from the target country of interest are available for model training.

Our multi-tiered methodology could probably be further improved by extra ground-truth surveys and/or more inputs, such as higher resolution or more frequent satellite imagery. However, we believe that the current availability of satellite imagery and state of deep-learning methods is more than adequate for making progress on a series of challenging questions related to the causes and consequences of infrastructure development, agriculture productivity improvements, public health interventions and broad-scale environmental change, and that their combination represents a new frontier for policy evaluation. Finally, our results point to the importance of carefully considering 'downstream' causal inference tasks when selecting an 'upstream' model that will feed data to those tasks. In the setting considered here, customary approaches to optimizing prediction performance can substantially worsen bias in the causal estimate of interest, a dynamic we expect exists in many related applications.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-022-05322-8.

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Methods

Dataset construction

Creating an asset-based wealth index. The DHS is a set of nationally representative household surveys conducted periodically in many low-income and middle-income countries around the world. Among other things, the DHS ask a relatively consistent set of questions on household asset ownership and related housing characteristics, including questions on water supply, ownership of assets such as a car and floor, roof and wall materials. Following earlier work[3], we convert categorical variables such as floor type to a 1–5 score indicating the quality of the asset. We then construct an asset wealth index at the household level through principal component analysis of survey responses to a consistent set of 12 asset-related questions (Extended Data Fig. 3). Following standard practice[33], we construct our asset wealth index from the first principal component of responses to these 12 questions; this first component explains 30% of the total variance in asset ownership in our sample. Regarding ownership of specific assets (such as radio ownership), the hypothesis proposed by the proponents of electrification argues that community electrification, among other things, will lead to new businesses being created and existing businesses becoming more productive, leading to wage and employment growth. If true, we should expect to see an increase in both consumption goods and goods used in household economic production. For this reason, we do not expect the effect of electrification to show up only for electrified goods.

Such an index is a standard approach in development economics to measuring longer-term well-being[21,33] and is meant to capture household asset ownership as a single dimension, rather than act as a direct measure of consumption poverty. By construction, the wealth index is normalized to have a mean equal to 0 and a standard deviation of 1 across all households, meaning that below average wealth will have negative values. Because our treated and control units are largely from rural Uganda, where wealth is typically lower than that in urban areas, the wealth index for those communities are, on average, below the mean and negative, as seen in Fig. 3a.

Included DHS data come from 25 countries in SSA between 2005 and 2017, totalling 641,621 households in 27,174 EAs (roughly villages in rural areas or neighbourhoods in urban areas) (Extended Data Fig. 2). In Uganda, we use five DHS (2006, 2009, 2011, 2014 and 2016), amounting to 1,798 EAs. EAs are geolocated in DHS; however, each reported location includes random jitter (spatial offset) to protect anonymity. Urban clusters include up to 2 km displacement and rural areas commonly up to 5 km and sometimes 10 km, which introduces random noise into our data setting but does not substantially affect predictive quality[2]. Before using the DHS data in model training or causal inference, we average household wealth indexes at the EA level. We aggregate to the cluster level because DHS data are not geolocated at the individual level, only at level of the EA.

Because of our focus on measuring the impact of electricity access, we do not include the DHS' binary 'has electricity' (HV206) survey question in our construction of the asset index to avoid mechanically embedding our independent variable of interest in our dependent variable. However, because ownership of individual household assets and housing characteristics tend to be highly correlated, excluding this input variable does not meaningfully change our wealth index. We compare our base wealth index to an index that includes the electricity variable, an index that has the binary electricity variable but not the electrical appliances and an index that has neither the electricity variable nor electrical appliances (Extended Data Fig. 4). We find high correlation between the wealth index we use and these alternative indexes, with r^2 values of 0.99, 0.97 and 0.96, respectively.

Constructing the electricity grid time series. To create the electricity grid time series, we collected data from a variety of publicly available sources, including published government documents, academic studies and aid agency reports that mapped electrical distribution system in Uganda. In total, we assembled Ugandan grid location data for the years 2005, 2006, 2010 and 2013 to 2018 and explicitly use data from 2010, 2013 and 2016 (refs. [34–36]). To verify the accuracy of these maps, we conducted in-person interviews in Kampala with utility professionals and aid agencies and contacted report authors.

Data from 2016 and later were made available in digital format from the Ugandan government, in collaboration with the World Bank and GIZ (the German Agency for International Cooperation). Older grid data were extracted from non-digitized maps through map rectification. We accomplished this task by uploading static images to ArcGIS, extracting grid lines and converting them to a digital latitude and longitude format. To further confirm the accuracy of the earlier, non-digitized maps, we developed a 'back-casting' strategy in which, under the assumption that, once the grid is built, it rarely disappears, we mapped the most recent years first and then used these maps to inform grid locations in earlier years. For example, with the 2016 grid mapped, we could overlay an earlier year and have the primary aim of tracking where the grid stopped in that year, rather than mapping that entire earlier year by itself. To our knowledge, our resulting grid expansion dataset is the first national-level time series of the evolution of an electric grid in SSA.

Identifying treatment and control groups. We denote communities (either DHS EA locations or villages from across Uganda) as 'treated' if they received distribution grid access between 2011 and 2012. Our treatment does not claim that all homes or businesses received and used grid-based electricity. Rather, our treatment assumes that the community received grid access. To identify our treatment and control groups, we apply a 2-km buffer to our electricity grid lines; that is, DHS locations or grid cells are considered treated if they fall within 2 km of the electrical grid constructed between 2010 and 2012 and are assigned either their DHS-derived wealth estimates or the CNN-derived wealth estimate for the CNN grid cell they fall within, as described below. (We test results when using a 3-km or 4-km buffer (Extended Data Fig. 8) and find similar treatment effects.) Using the time series of grid development, we create a list of EAs (or villages in the full-country setting) that were connected pre-2011, during 2011 and 2012 or post-2012. We define control locations as sites that were not within 2 km of the electrical grid by 2016; these villages are similarly assigned their DHS-derived wealth estimates or the CNN-derived wealth estimate for the CNN grid cell they fall within.

In the full-country setting in which we apply the CNN across Uganda rather than just in DHS locations, we modestly update our treatment-selection methodology. First, because there is no random jitter in village coordinates, we shrink our treatment range to 1 km on either side of the grid line. Second, as many villages are tightly clustered in this data setting, we add a 1-km buffer outside the 1-km treatment range, meaning that all control units are still beyond 2 km from the grid line. In this setting, we end up with 268 treated villages and 3,601 control villages, and predict the wealth for each annually from 2006 to 2016.

Wealth prediction

Satellite-based wealth estimation in the context of downstream inference tasks. We wish to infer the impact of electrification on livelihoods, using data on livelihoods predicted by a model. For continuous outcomes, the distribution of predicted outcomes from both simple linear models and more complex deep-learning models tend to have lower variance than the distribution of observed outcomes. This is mechanically true for linear regression models and is empirically true for deep-learning models in our setting (Extended Data Fig. 5a). In practice, this means that predicted wealth is overstated for poorer segments of the population and understated for wealthier segments of the population, meaning that prediction bias is a function of underlying wealth.

We explore whether this so-called Berkson error can be consequential for downstream causal inference tasks in our setting. Consider a

standard two-period DiD estimation, with true wealth values y measured in both periods for treatment and control units, and predicted (potentially biased) wealth values y' in the same units and periods. Assume that predicted wealth is a linear transformation of true wealth, that is, $y' = \alpha + \phi y$, in which, in general, we expect $\alpha > 0$ and $0 < \phi < 1$. Predicted wealth is too high at low wealth levels and too low at high wealth levels.

For notational simplicity, we label period and group-specific true and observed wealth values as T_1 and T_1' for true and predicted wealth in the treatment group in the treated period, and do not take expectations. (We use this simplified notation for the sake of clarity; strictly speaking, T_1 refers to the expected value of treated units in the post-treatment period observed without error, that is, $T_1 = \mathrm{E}[y|D = 1, T = 2]$. Likewise, T_1' is the same quantity but observed with measurement error: $T_1' = \mathrm{E}[y'|D = 1, T = 2]$). The 'true' DiD estimate is given by:

$$\beta = (T_1 - T_0) - (C_1 - C_0) \tag{1}$$

The DiD estimated on predicted data is:

$$\beta' = (T_1' - T_0') - (C_1' - C_0') \tag{2}$$

Using the mapping from true to predicted wealth, we can then write the DiD bias as:

$$\begin{aligned} \beta' - \beta &= [(T_1' - T_0') - (C_1' - C_0')] - [(T_1 - T_0) - (C_1 - C_0)] \\ &= [((\alpha + \phi T_1') - (\alpha + \phi T_0')) \\ &\quad - ((\alpha + \phi C_1') - (\alpha + \phi C_0'))] \\ &= -[(T_1 - T_0) - (C_1 - C_0)] \\ &= [\phi(T_1 - T_0) - \phi(C_1 - C_0)] - [(T_1 - T_0) - (C_1 - C_0)] \\ &= (\phi - 1)\beta \end{aligned} \tag{3}$$

Bias is only a function of the slope parameter ϕ and not intercept α (any level bias nets out in DiD). Because we expect $0 < \phi < 1$, then our DiD estimate will be attenuated: downward biased if the true effect is positive and upward biased if the true effect is negative in our two-period setting. We confirm using simulation that Berkson error also leads to attenuation bias in our MC estimator (Extended Data Fig. 6c). We thus seek model predictions in which ϕ is as near to 1 as possible, that is, the Berkson error is eliminated.

Wealth estimation with a CNN. We seek a prediction model that can accurately predict local-level wealth using inputs that are available comprehensively across space and time, and whose predictions do not suffer from the Berkson-type error described above. Coarse-resolution public satellite imagery from Landsat and other sensors is one such available input and previous work has shown that a CNN can be trained to accurately predict local-level asset wealth using Landsat as input[2,3]. However, the use of standard CNN loss functions that minimize MSE across the entire output distribution tend to generate wealth predictions with the Berkson-type error described above.

We thus develop and embed a custom loss function in a ResNet-18 CNN architecture that is able to generate wealth predictions from satellite inputs without Berkson error. Our basic approach is to also penalize bias at separate quintiles of the wealth distribution. This reduced bias comes at the expense of overall predictive performance (for example, as measured by r^2) and, generally speaking, the best models on r^2 grounds tend to also produce the largest Berkson error. This suggests that researchers wishing to use data generated from CNN and other ML architectures should tailor their loss function according to their desired downstream inference task.

We use a ResNet-18 architecture[23] with a modified input layer to accommodate our multispectral input data (224 × 224 × 6) and a modified final layer to produce a single continuous estimate instead of predictions for several classes. The model is trained from scratch using an Adam optimizer[37] with a learning rate decay of 0.96 per epoch. We split our data into five equally sized subsets and set up five different random folds, in which three of the subsets (60%) are assigned to the training set and one subset (20%) is assigned to the validation and test sets each. The validation data are used to tune the model, whereas final performance statistics are calculated using the test set. We ensure that all five subsets are used as the validation or test set exactly once.

Custom loss function. We define the five quintiles of the survey-derived wealth distribution as Q_1 to Q_5, with Q_1 being the lowest quintile of wealth. Consider a mini-batch (a subset of the data used in each training step) with K training data points X_i with wealth labels y_i. The CNN at this current step can be defined as $\hat{f}$ and the predicted wealth is given by $\hat{y} = \hat{f}(X_i)$. We seek to compute the bias B of the estimator $\hat{f}$ with respect to each quintile of wealth and then penalize some aggregation of this bias. Ideally, we would measure $B_1(\hat{f}), \ldots, B_5(\hat{f})$ as the bias that $(\hat{f})$ generates over all the data points in Q_1 to Q_5.

However, in practice, we are constrained to using the data in each mini-batch and instead compute the sample bias $\hat{B}_1(\hat{f}), \ldots, \hat{B}_5(\hat{f})$ for each quintile as follows:

$$\hat{B}_j(\hat{f}) = \mathbb{E}[\hat{f}(X_i) - y_i \,|\, y_i \in Q_j] \tag{4}$$

Using these quintile-specific bias estimates, we then define our custom loss function:

$$\hat{E}_b = \max_j(\hat{B}_j(\hat{f})^2) \tag{5}$$

with $\hat{B}_j(\hat{f})$ the estimate of the true bias of the CNN on the jth quintile $B_j(\hat{f})$. This estimation of the bias will hence have an error associated with it that becomes smaller as the number of samples fed to it (the y_i and $\hat{f}(X_i)$) increases. This is why we increase the batch size to 90, to allow for an average of 18 samples per quintile. Equation (5) forces the model to minimize the maximum quintile-specific bias.

We then combine this quintile-specific loss function with the standard MSE loss and regularization loss. We write the total loss equation for a given mini-batch as:

$$\mathcal{L} = \mathrm{MSE} + \lambda_r L_2 + \lambda_b \hat{E}_b \tag{6}$$

in which L_2 represents the L2 norm of the CNN weights (a standard practice to avoid overfitting). Finding the optimal value of λ_b requires a different approach, as it seeks to lower a specific type of bias at the cost of overall MSE, so any search based simply on the standard MSE metric would have returned $\lambda_b = 0$. Instead, we choose λ_b empirically to yield a ϕ (the slope of the regression of true wealth on predicted wealth) as close to 1 as possible. Extended Data Fig. 5 demonstrates the impact of using no quintile-specific bias ($\lambda_b = 0$) versus increasingly weighting the quintile-specific bias term (λ_b up to 7.5). Empirically, we find $\lambda_b = 5$ to yield predictions with ϕ closest to 1 (Extended Data Fig. 5d), so we choose this penalty as our primary model. We again note that this reduced bias comes at the cost of an overall reduction in prediction r^2 (Extended Data Fig. 5f), relative to a model with a standard loss function ($\lambda_b = 0$).

Our final CNN parameters are an initial learning rate of 0.0001, a regularization term of 0.0001 and a bias penalty term of 5. Using these model parameters, we produce five disjoint splits that estimate a wealth index for every DHS location or grid cell in Uganda on an annual basis from 2006 to 2016. With five splits, this means that every year-unit observation will have five independent estimates.

Imagery inputs. Earlier efforts to predict local-level well-being from imagery have typically used a combination of daytime and nighttime

imagery[3,4]. In our analysis, we use only daytime imagery from Landsat as input, as we are concerned that nighttime light imagery was probably a direct proxy for electrification and, thus, its inclusion (much like the inclusion of household electrification in the asset index) could mechanically embed our independent variable of interest into our dependent variable of interest.

We use six bands in Landsat: the standard three RGB visual bands, one near-infrared band and two shortwave infrared bands. All of the bands have a native resolution of 30 m per pixel and so our standard CNN input patch size of 224 pixels means that each input image approximately covered a 6.72×6.72 km^2 area. Our goal is to obtain data for each pixel in each year. As Landsat has many images at a given location for a year, we generated a composite by taking the median pixel value over all non-cloudy images falling within the year span. Taking a composite in such a way enables us to operate on a single image (as opposed to the more complex case of a sequence of images) and conceptually removes confounding seasonal weather effects or other scene-specific artefacts. Compositing was done in Google Earth Engine.

Training with and without Uganda. Although our target geography is Uganda, we have a wealth of data from dozens of other African countries on which to train our CNN. Past studies have suggested that models trained in one African country tend to perform well in nearby countries[3,4] and that—for daytime-imagery-based models in particular—having many training observations improves model performance[3].

We thus train our CNN using data from Uganda as well as from other African countries. In our core 'with-Uganda' scenario, we also allow observed DHS clusters from Uganda to be randomly selected into the CNN training and validation sets, which—intuitively—could increase performance in our target geography of interest. We also create a 'without-Uganda' setting in which we do not allow the CNN to see any Uganda data in the training and validation sets. This is designed to replicate a challenging but common real-world situation in which target geographies may have no data for training or validation. Our preferred bias-corrected CNN models in the with-Uganda case have an average r^2 of 0.64 across SSA and that for the without-Uganda case is 0.63.

In our 'full-country' setting, we use the same CNN parameters as our base with-Uganda case (initial learning rate of 0.0001, a regularization term of 0.0001 and a bias penalty term of 5) and allow the CNN to train on Ugandan DHS clusters. We then impute wealth indexes across all of Uganda from 2006 to 2016 (Fig. 2a,b). Next, we use a geolocated data frame of Ugandan villages and neighbourhoods[38] and extract wealth indexes for each village on an annual basis; this ensures that we are extracting estimates for populated areas. We ultimately end up with about 6,900 village estimates per year (Fig. 2c).

Our deep-learning models probably extract a range of information from input imagery that is useful for predicting variation in local-level livelihoods. Although in this study we do not interrogate why our model is capable of making accurate wealth predictions (instead convincing ourselves that this is so by evaluating predictions on data on which the model was not trained), related work[3] using similar Landsat inputs visualized model-derived features to help understand what information the model was using. This work found that the model seemed to learn semantically meaningful features related to urban areas, agricultural regions, water bodies and the presence of deserts.

Causal inference

We explore the use of three different causal inference techniques, the well-known DiD estimator and two ML approaches, MC and SC-EN[24,25]. We evaluate their appropriateness in various data settings.

DiD. DiD estimates the impact of a treatment by subtracting changes in outcomes in an untreated group from changes in outcomes from a newly treated group. Most DiD estimators are used in a panel setting in which both control and (eventually) treated groups are repeatedly observed. In the DHS survey data, the sampling frame changes between survey rounds, meaning that the same villages are not observed in separate surveys. However, in the absence of panel data, DiD could still be implemented in two ways on these survey data and provide a relevant benchmark against which to understand other estimators and results. First, we can simply compute the DiD estimator over the observed villages in each round, defining in each period the control villages that were never treated and the treated villages that were eventually treated, and computing mean outcomes over each group. We then stack the control and treated averages across periods and implement DiD as a two-way fixed-effects regression:

$$Y_{it} = \beta\tau_{it} + u_i + v_t + \varepsilon_{it} \tag{7}$$

in which Y_{it} is the cluster-level wealth index, τ is a binary indicator for whether the grid was present, u are unit fixed effects, v are time fixed effects and ε captures the remaining error. Reported estimates of β are shown as '2-unit DiD' in Fig. 3b.

A second approach is to spatially interpolate survey observations in each survey round and use interpolated values to create a village-level panel. To implement this approach, we take all 1,798 cluster locations that were observed in any of the five DHS surveys in Uganda (2006, 2009, 2011, 2014 and 2016) and in survey years in which that location was unobserved, we impute the wealth value in that location-year as the IDW average of wealth values in nearby villages, using all observations within 10 km of the location in the average; locations without observations within 10 km are dropped. We then run our standard two-way fixed-effects DiD regression, clustering standard errors at the EA level ('IDW DiD' in Fig. 3b). We note that our DiD setting does not have staggered adoption and only has never-treated units as controls, and thus avoids the issues associated with DiD highlighted by several recent papers[39,40].

Testing parallel trends. Causal identification in DiD requires a parallel-trends assumption, in other words, that outcomes in control and treatment groups would have trended similarly in the absence of treatment[19,20]. A standard check on this assumption is to evaluate whether outcomes were evolving similarly before treatment. To run this test, we use panel data imputed by our CNN model. We then conduct each test by relabelling treated units in 2010, our last pre-treatment year, as being treated in that year and run the above DiD regression using data up to 2010. If the DiD estimator is statistically different from zero (P-value < 0.05 at 95% CI), we reject parallel pre-trends. We conduct this parallel-trends test on the same with- and without-Uganda samples, including randomized CNN split values, as we use in the final causal analysis. The pre-trends tests failed slightly more than would be expected by chance in the setting in which Uganda data were used in training and substantially more than would be expected when Uganda data were not used in training (Extended Data Fig. 12).

SC-EN and MC. SC-EN[25] is a regularized form of the synthetic control method[41], which uses elastic net (a combination of LASSO and ridge penalties) to limit overfitting when matching treated units to control units. In general, SC-EN is a more flexible model than traditional synthetic controls because it allows weights to be negative and does not require them to sum to zero: formally, SC-EN estimates the unit specific counterfactual as:

$$\hat{Y}_{j,T}(0) = \hat{\mu}^{\text{en}}(j;\alpha,\lambda) + \sum_i^{\text{en}} \hat{\omega}(j;\alpha,\lambda) * Y_{i,T}^{\text{obs}}. \tag{8}$$

in which $\hat{Y}$ is the counterfactual being predicted, $\hat{\mu}$ is an intercept, α is how much weight to put on the LASSO or ridge components of the elastic net estimator, which we fix at 0.5, λ weights the penalty function and is determined by means of cross-validation, in which we find the

tuning parameter, λ, that minimizes the error in cross-validation tests[25], and ω is a weight for each control observation, $Y_{i,T}^{\text{obs}}$. In words, SC-EN predicts the counterfactual, $Y_{\text{cf,post}}$, by weighting each post-period control variable with ω_i, noting that zero weights are common. These weights are determined through panel-like regression within the pre-treatment period, in which we are regressing a single treated unit on the full panel of control units. In this setting, the control weights are revised for each treated unit.

In our setting, we have fewer time periods (T) than we do units/villages (N), so we run a transposed or perpendicular version of SC-EN following ref. [24]. Thus, we are regressing individual post-treatment control years on the panel of pre-treatment controls and estimating year weights, rather than unit weights. We apply these year weights to each pre-treatment treated-unit time series to estimate the year-specific counterfactual of each unit.

As in SC-EN, MC[24] attempts to impute (unobserved) untreated outcome values for treated units, but does so by treating these unobserved outcomes as missing values in a matrix to be imputed. Imputation is performed by estimating a parsimonious, regularized representation $\hat{L}$ of the original outcome matrix Y_{it} that minimizes the distance between estimated and observed outcomes. The algorithm penalizes matrix complexity by means of a regularization term λ chosen through cross-validation. Formally, the algorithm solves the following minimization problem:

$$\hat{\mathbf{L}} = \text{argmin}_L \left[\sum_{(i,t)\in\vartheta} \frac{(Y_{it} - L_{it})^2}{|\vartheta|} + \lambda ||\mathbf{L}||_* \right] \quad (9)$$

The completed (imputed) missing values in $\hat{L}$ are then the estimate of what the untreated outcomes would have been in the treated group, and the treatment effect estimated the average difference between the observed (treated) outcomes and the imputed untreated outcomes.

Whereas DiD requires that treated and control units be trending similarly in the absence of treatment, and is thus biased in the presence of time-trending unobservables, both MC and SC-EN only require that some subset of control units have trends that are correlated with the trends of a treated unit. These newer estimators are thus, in principle, more robust to pre-trends violations and have been demonstrated to perform well in a range of prediction settings[24] (Extended Data Fig. 7). Notably, the approach retains the ability to use both time and unit fixed effects.

Estimator validation. We performed a series of tests on SC-EN and MC to evaluate their performance in our setting. First, we used k-fold cross-validation to evaluate the precision of the SC-EN and MC models and their performance against DiD in predicting held-out post-treatment values in the control set. We accomplished this by splitting the Uganda control villages into ten random, equally sized partitions, repeatedly holding out one fold, setting post-2010 values to missing for villages in this fold, and using the remaining 90% of control observations to impute these missing values using SC-EN, MC or DiD for prediction. Every fold was used as a test set one time and in the training set $k-1$ times, resulting in a unique yearly estimate for each control unit in the post-treatment period. We then compared each observed control unit value (y_{it}^{c}) to each predicted control unit ($\hat{y}_{it}^{c}$) and computed the root-mean-square error. We also compared the annual observed mean to the annual imputed mean, which reflects average model accuracy, our primary goal. We found that, across all post-treatment years, the mean difference is 0.013, 0.017 and 0.018 for DiD, MC and SC-EN, respectively. The root-mean-square errors are 0.36, 0.45, and 0.44, respectively. Cross-validation results are shown in Extended Data Fig. 6.

Second, we assessed how well DiD, MC and SC-EN predicted outcomes for treated units in 2010, the last pre-treatment year. If outcomes were trending differentially between treated and control groups before treatment, then an ability to accurately predict outcomes in the control group (our first test above) would not guarantee an ability to predict counterfactual outcomes in the treated group. To evaluate this latter ability, we randomly selected control and treatment units, with replacement, into a sample dataset. To assess DiD, we ran a standard pre-trends test and computed the change over time in pre-treatment outcomes in the treated group minus that in the control group. To assess MC and SC-EN, we predicted outcome values for treated units in 2010—the last pre-treatment year—and compared predicted to observed values. We ran this analysis 100 times on a random selection of control and treated units and averaged the results. We find that MC and SC-EN perform similarly to DiD in terms of average prediction error (−0.03, -0.03, 0.03), despite a limited set of pre-treatment years that can be used to train.

Finally, we explore the impact of parallel-trends violations on our different estimators. Non-parallel pre-trends are a plausible concern in our data—for example, the concern that electrification is targeted to locations that were already growing most quickly—and, as shown in the above DiD tests, is perhaps a practical concern as well. We explore a simulated multi-year setting in which treatment has a positive effect (equal to 1) but treated units pre-treatment are, on average, trending positively relative to control units (that is, parallel trends is violated). We then estimate treatment effects using DiD and MC, varying the number of years in the sample (t = even years between 4 and 22), in which treatment can occur at the earliest in year $t/2+1$, and varying the magnitude of pre-trends violation. We found that MC outperforms DiD in most scenarios (Extended Data Fig. 7), particularly as pre-trends violations increase or the sample exceeds a few years.

Overall, we find MC to outperform DiD in two of our robustness checks and perform similarly in a third. Recognizing the other literature highlighting the performance of MC[24], we select MC as our preferred causal estimator. We also report SC-EN in our results because it performs well in the most common cross-validation test.

Estimating the average treatment effect

To calculate average treatment effects and their uncertainty, we use a common bootstrap procedure in which we randomly select control and treated units with replacement into a sample dataset and re-estimate each of our causal models (MC, SC-EN or DiD). To capture imputation uncertainty, every bootstrap run starts by randomly selecting a single unit-year wealth index for each of our control and treatment units (recall that our CNN procedure splits our training data into five folds and, thus, produces five unit-year estimates for each location). Using our base 2-km setup as an example, we then randomly select 888 control and 76 treated units to create a run-specific sample dataset. Next, we estimate MC and SC-EN, recording each estimated counterfactual for treated units from 2011 to 2016. As MC re-estimates the entire sample dataset, we get all of the estimated counterfactuals in one matrix. For SC-EN, we run our algorithm separately for each post-treatment year. Then, to estimate the year-unit specific treatment effect, we subtract the estimated counterfactual from the observed value. To estimate annual average treatment effects, we average the within-year estimates. We repeat this analysis on 500 distinct datasets for our main results and report bootstrap standard errors at the 95% level of confidence for both MC and SC-EN (Fig. 3). We limit our analysis to 100 model runs for 3-km and 4-km results (Extended Data Fig. 8), as well as the 0, 1, 3 and 7.5 penalty results (Extended Data Fig. 9) for computational efficiency.

Our core results—for both the with-Uganda and the without-Uganda cases—are based on a 2-km buffer around the electric grid locations and use a penalty term of 5 on the quintile-specific bias. We also report 3-km and 4-km buffer results using the same penalty term of 5 and find similar results (Extended Data Fig. 8). And we estimate results when the penalty term is set at 0, 1, 3 and 7.5 (Extended Data Fig. 9). Estimates are larger for penalty terms that minimize Berkson error in the wealth predictions.

To understand whether the expansion of the electrical grid was correlated with other public or private investments that might have independently affected local livelihoods, we assembled and harmonized data on road construction, investments in road improvement, cell phone ownership and cell phone coverage. For road construction, we use data from ref. [26], which track road development across SSA beginning in 1960. Ref. [26] provides road data for 23 years in Uganda, with two years occurring during our study period, 2007 (pre-treatment) and 2012 (post-treatment). During this 6-year period, their data indicate that no new major roads were added, suggesting that new road construction did not occur in tandem with electricity grid extension, including during our treatment period (2011–2012). To study road improvements, we digitized reports of road-improvement projects from the Uganda National Roads Authority, which provided text-based reports of road-improvement projects. Between 2010/2011 and 2015/2016, road upgrade projects averaged 184 km per year, totalling 1,102 km during the period—which is less than 1% of the road network in Uganda. We compared the spatial extent of road improvements relative to electrification expansion, finding no overlap between the two (Extended Data Fig. 10).

For cell phone coverage, we used household mobile phone ownership data from Ugandan DHS rounds (2006, 2009, 2011, 2014 and 2016) and cellular phone coverage and quality from GSMA (2010–2016). To test whether grid expansion was correlated with mobile phone ownership, we ran a DiD test with phone ownership as the outcome and electrification as the independent variable, as well as an accompanying pre-trends test. Before electrification, we find no evidence that cell phone ownership was differentially trending in un-electrified versus soon-to-be electrified villages ($P = 0.11$ on the null of no effect). We then estimated that electrification expansion during our study period reduced mobile phone ownership by −0.015 percentage points (Extended Data Fig. 11a), suggesting that ownership grew faster in control communities post-electrification expansion. This finding is consistent with lower-income regions catching up to wealthier regions in terms of cell phone ownership during our study period.

Using maps of cell phone coverage and coverage quality from GSMA, we run similar DiD tests. Here quality of coverage is defined by GSMA as non-existent (NA), strong (1) or variable (2). Some data anomalies were found (7% of the data in our sample), in which values between 1 and 2 were listed. We leave these input values in the data frame because GSMA itself recognizes that some small data-entry errors were made by cell service providers, but also leaves the data in the set. To make inference clearer, we recode the data to 0, 1 and x/4, respectively. We find similar negative effects, with electrification associated with an apparent decline of −0.098 percentage points in coverage and −0.07 in quality of coverage, respectively (Extended Data Fig. 11b,c). This again indicates that coverage was growing faster in control areas, which are poorer at baseline. Taken together, our results indicate that expansion of road or cell-phone networks were uncorrelated with electrification expansion during our study period, and thus that our main results are unlikely to conflate the impacts of electrification with the possible impacts of these other investments.

Finally, to understand how electrification might have affected the ownership of specific assets contained in our asset index, we assessed changes in asset ownership separately for 11 tangible assets that contributed to the construction of our wealth index. Because these survey data are cross sectional from different rounds of DHS, we average responses to make a two-unit (control and treatment), two-period (pre and post) panel. We then run a DiD test and report the results in Extended Data Fig. 13, comparable with the two-period DiD reported in Fig. 3b. We find statistically significant positive effects for three home-materials-related variables, changes in which could plausibly be identified in imagery. We also find positive results in several electricity-related variables, indicating that homeowners are experiencing increased levels of electrification. However, we find no substantively or statistically meaningful effects for more expensive products, such as cars and motorcycles.

Research-related carbon emissions

We recognize that deep learning requires large amounts of computational power and energy, typically resulting in CO_2 emissions from energy use. At the same time, ML can save emissions if it offsets more consumptive data collection strategies. Although the artificial intelligence community should be keenly aware of its energy consumption and emissions, we argue that, in our setting, there is probably a net benefit to generating data on economic livelihoods using ML algorithms, as ML-based predictions can offset or augment in-person data collection, as is done in DHS and other livelihood surveys. For example, the Ugandan DHS we use include 1,798 EAs across 5 years, whereas our imputed data fills in gaps to produce a net of 19,778 EAs across 11 years, more than ten times as much data. Thus, assuming that some reasonable level of further in-person-based surveys would be required to conduct a similar analysis (without using ML), it is probable that our ML approach will produce lower total emissions than a counterfactual in which further field data needed to be collected. In our setting, as the heaviest computational work was done on the cloud by means of Google's data centres, our total emissions profile benefits from Google's progressive clean-energy commitments, as well as those of Stanford University. Increasing clean-energy penetration will further this advantage and reduce the ecological footprint of the ML community.

Data availability

Data and R code to replicate all the main results are available on GitHub at https://github.com/nwrat/RCQSB2022_public.

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Author contributions N.R. conceived the study, collected the data, led the econometric analysis and wrote the paper. G.C. designed and performed the CNN modelling, developed the bias penalty term and wrote the paper. B.d.l.C. contributed to econometric analysis and wrote the paper. M.S. contributed to the econometric analysis and wrote the paper. M.B. advised the project, contributed to econometric analysis and wrote the paper.

Competing interests M.B. is a cofounder at Atlas AI, a company that uses machine learning to measure economic outcomes in the developing world. G.C. is an employee at Atlas AI.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41586-022-05322-8.

Correspondence and requests for materials should be addressed to Marshall Burke.

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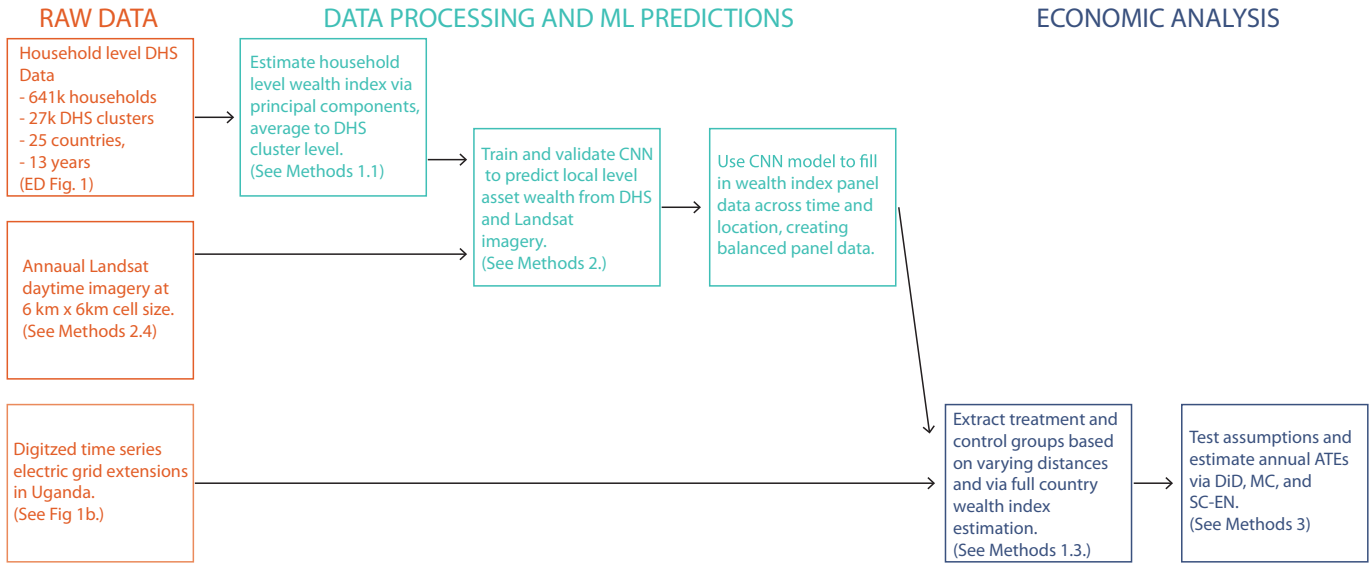


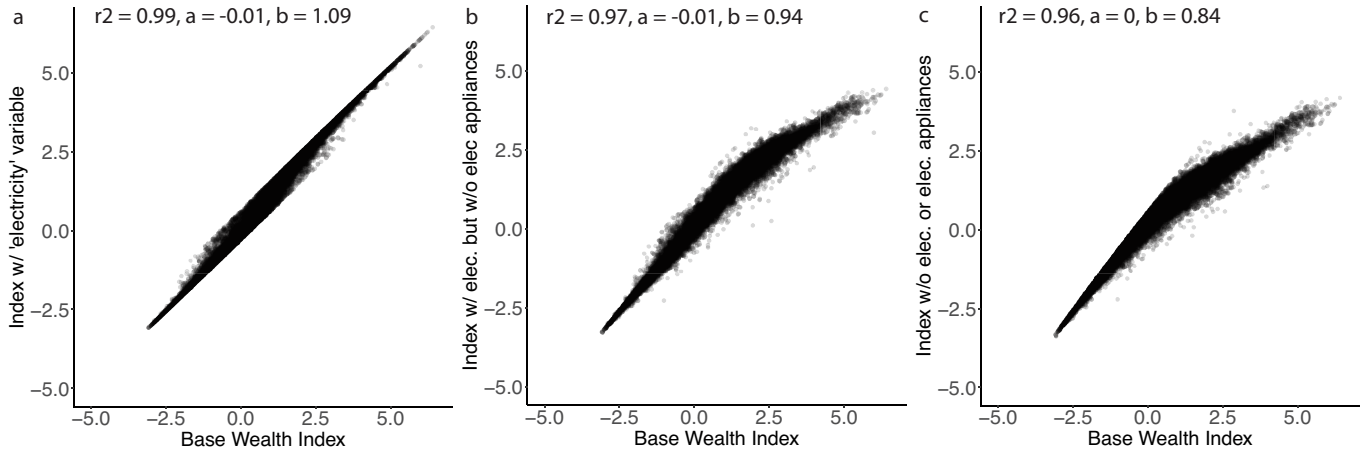
Extended Data Fig. 1 | Data and analysis pipeline. Starting with three forms of raw data on the left, a CNN was trained and modified to predict annual estimates of community-level asset wealth with reduced bias across 25 countries and 27,174 communities in SSA. Two ML approaches (MC and SC-EN) were then used to predict unit-specific unobserved counterfactuals for locations that received electrification between 2010 and 2012.

	Country	Survey Year	# Villages	# Households
1	Angola	2015	625	15110
2	Benin	2012	750	11322
3	Benin	2017	540	6223
4	Burkina Faso	2010	540	17304
5	Burkina Faso	2014	248	13717
6	Zambia	2013	376	8479
7	Zimbabwe	2005	194	3380
8	Zimbabwe	2010	552	15853
9	Burundi	2012	492	16354
10	Zimbabwe	2015	341	9078
11	Burundi	2010	576	12152
12	Burundi	2016	571	15794
13	DR Congo	2013	321	12271
14	Cote D'Ivoire	2012	404	9619
15	Cameroon	2011	612	3920
16	Ethiopia	2011	192	4758
17	Ethiopia	2016	300	7064
18	Ghana	2008	397	8018
19	Ghana	2014	1557	29423
20	Ghana	2016	245	5877
21	Guinea	2012	395	8954
22	Kenya	2008	399	9179
23	Mali	2015	886	7843
24	Malawi	2010	148	4222
25	Kenya	2014	827	23963
26	Kenya	2015	140	3401
27	Lesotho	2009	140	3399
28	Lesotho	2014	850	26226
29	Mali	2012	306	3655
30	Malawi	2012	409	5920
31	Malawi	2014	177	13919
32	Malawi	2015	270	7144
33	Malawi	2017	610	33312
34	Mozambique	2009	239	5359
35	Mozambique	2011	889	37416
36	Mozambique	2015	322	7463
37	Uganda	2016	479	5357
38	Zambia	2007	509	6214
39	Nigeria	2008	385	8155
40	Nigeria	2010	200	12550
41	Namibia	2013	492	4571
42	Rwanda	2010	435	12114
43	Rwanda	2015	336	6601
44	Nigeria	2013	214	7718
45	Nigeria	2015	344	4159
46	Namibia	2006	481	4496
47	Senegal	2012	458	7884
48	Sierra Leone	2008	608	8289
49	Sierra Leone	2013	329	9233
50	Sierra Leone	2016	336	9733
51	Senegal	2010	466	12559
52	Senegal	2015	573	7944
53	Togo	2013	319	4321
54	Tanzania	2007	170	8868
55	Tanzania	2010	400	5024
56	Tanzania	2012	208	17217
57	Tanzania	2015	684	7077
58	Uganda	2006	719	15546
59	Uganda	2009	396	9141
60	Uganda	2011	393	9324
61	Uganda	2014	400	10435
62	Total	NA	27174	641621

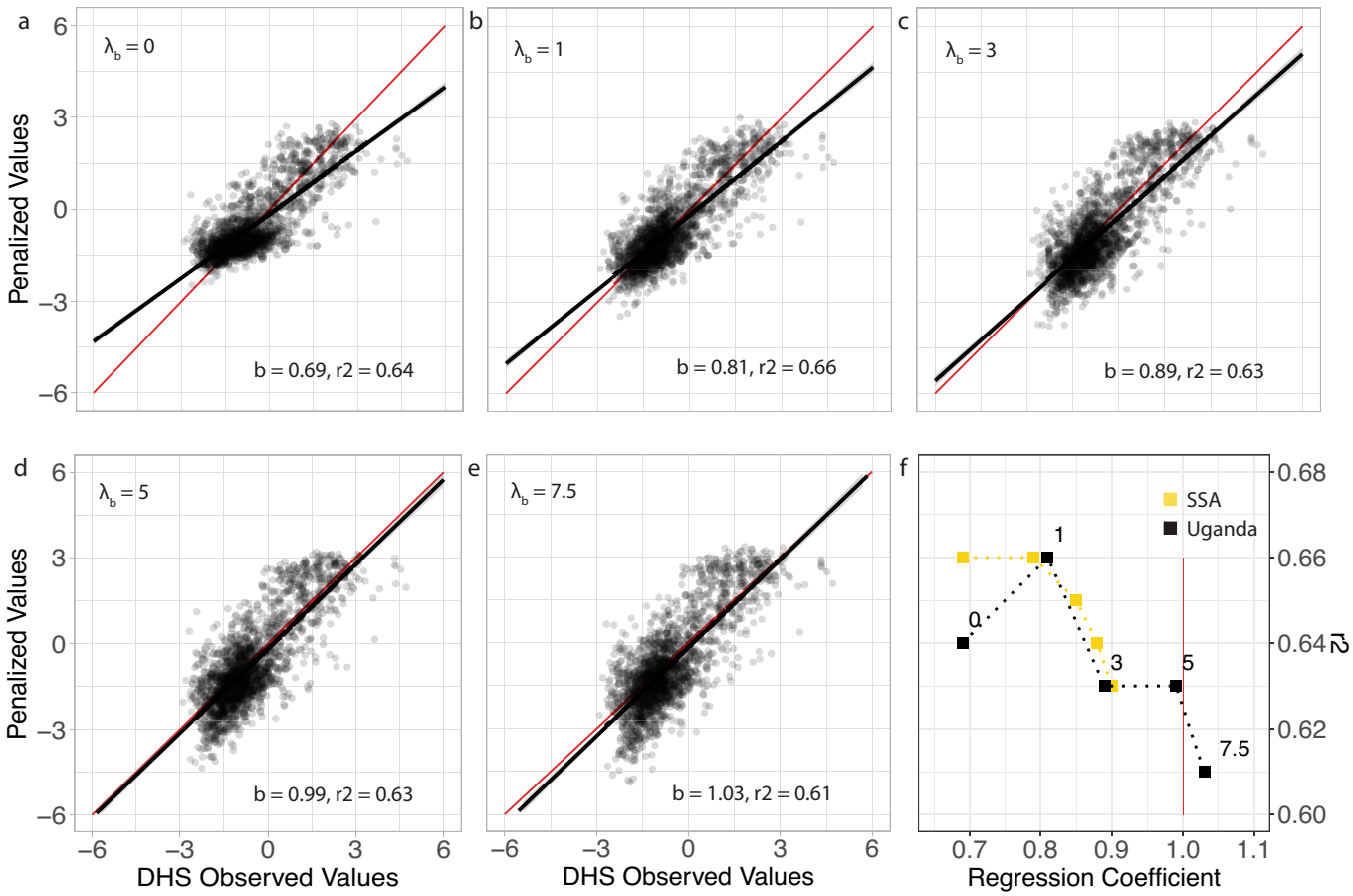
Extended Data Fig. 2 | DHS from SSA used in this study. DHS years in our sample run from 2005 to 2017. We used asset wealth estimates from more than 640,000 households across 27,174 villages to train our CNN wealth-prediction models.

	Variable	DHS Code
1	Drinking water source	HV201
2	Toilet type	HV205
3	Owns radio	HV207
4	Owns TV	HV208
5	Owns refrigerator	HV209
6	Owns motorcycle	HV211
7	Owns car	HV212
8	Primary floor material	HV213
9	Primary wall material	HV214
10	Primary roof material	HV215
11	Owns telephone	HV221
12	People per room	HV036 / HV 216

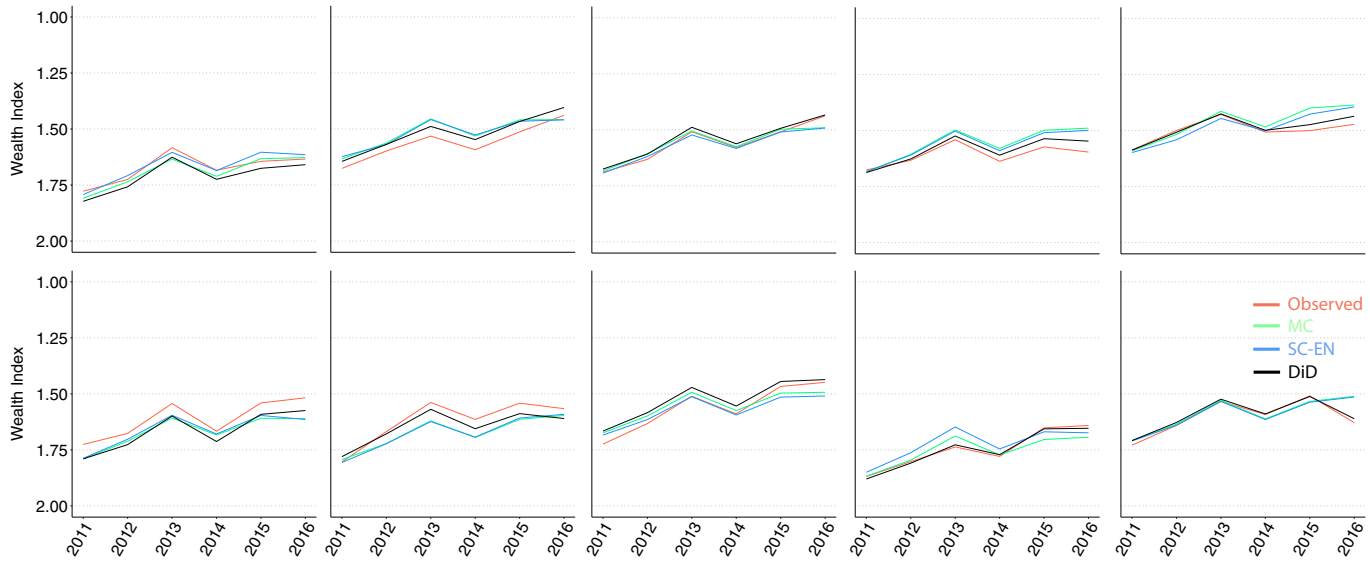
Extended Data Fig. 3 | Survey variables in DHS used in the creation of an asset wealth index in the training data. The column on the right gives the variable code used in the original DHS data.



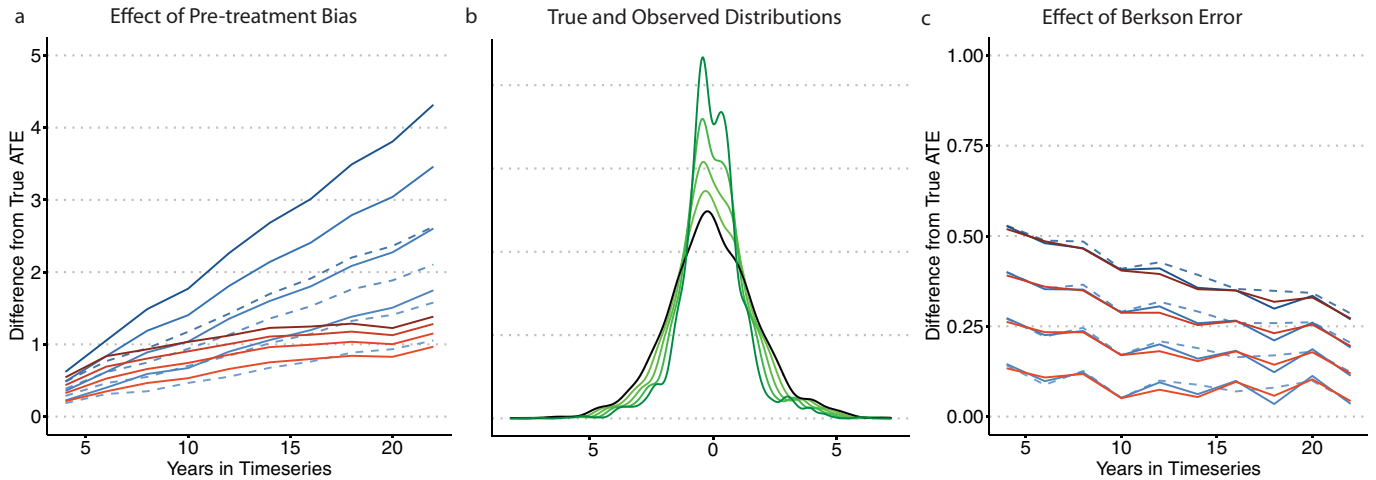
Extended Data Fig. 4 | Alternative approaches to constructing the asset wealth index result in indices that are highly correlated. a, Compares the base wealth index constructed from the variables listed in Extended Data Fig. 3 to an index that also includes the DHS' 'has electricity' variable. The r^2, intercept and slope coefficient from regressing the alternative index on the base index are shown at the top. **b**, Compares the base index to the base index plus the electricity variable but minus three electrical appliances (TV, refrigerator and phone). **c**, Compares the base index to the base index minus three electrical appliances.



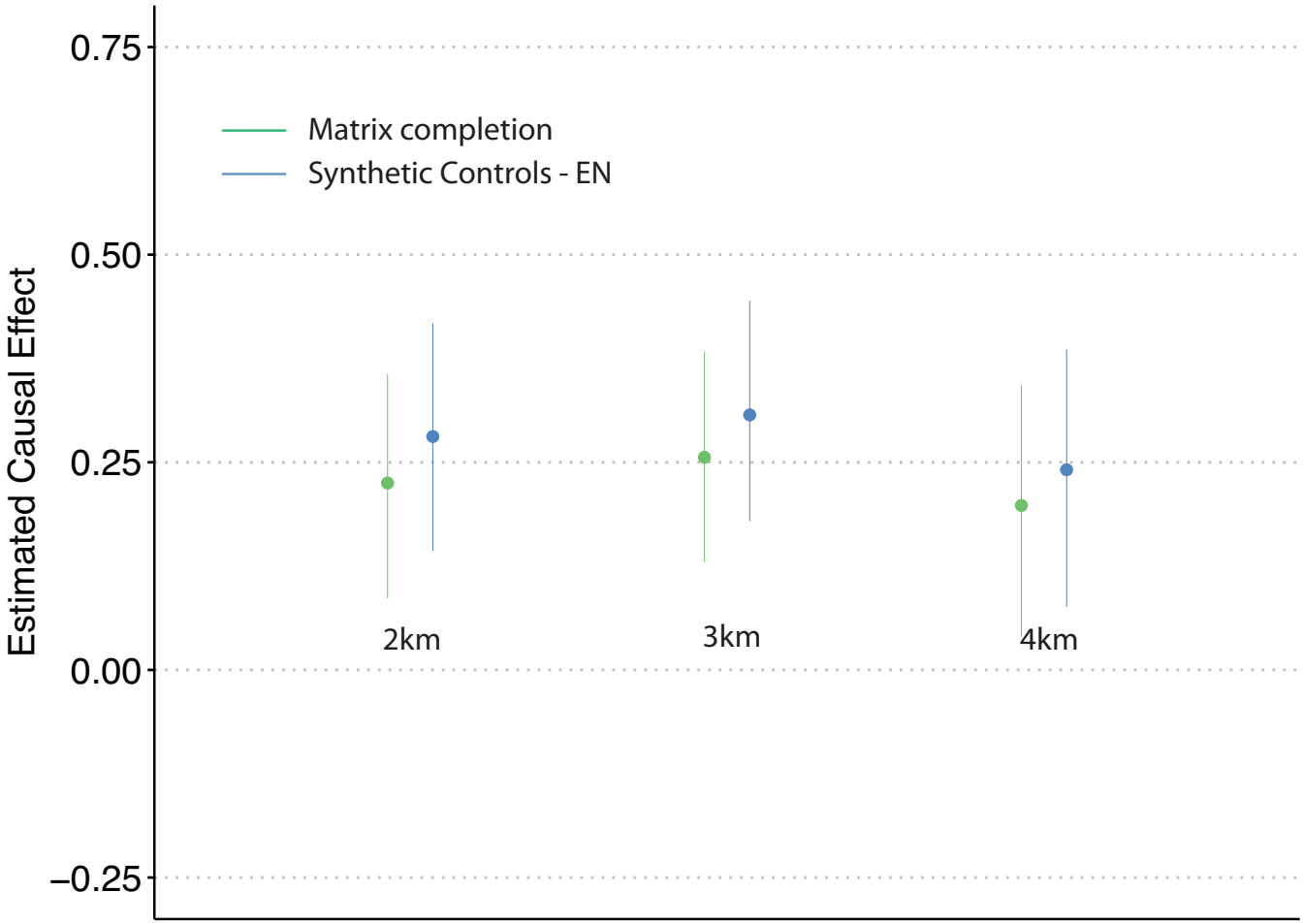
Extended Data Fig. 5 | Our custom loss function corrects for prediction biases at different points in the wealth distribution and modestly reduces overall predictive performance (r^2) in held-out Uganda data. a–e, DHS observed wealth index values in Uganda on the *x* axis ($n = 1{,}798$) and CNN-predicted values on the *y* axis. The red line is a 45° line, the black is the line of best fit from regressing predicted on observed; slope coefficient and regression r^2 are shown in the lower-right corner of each subplot. Higher penalties on quintile-specific bias leads to slope coefficients closer to 1 and slight reductions in r^2. However, when the penalty term gets too large, as in the 7.5 case in **e**, regression coefficients begin to depart from 1 again. **f**, Relationship between the slope coefficient and r^2 for each subplot. The results from the entire sub-Saharan dataset are shown in gold and those for the Uganda-only sample in black.



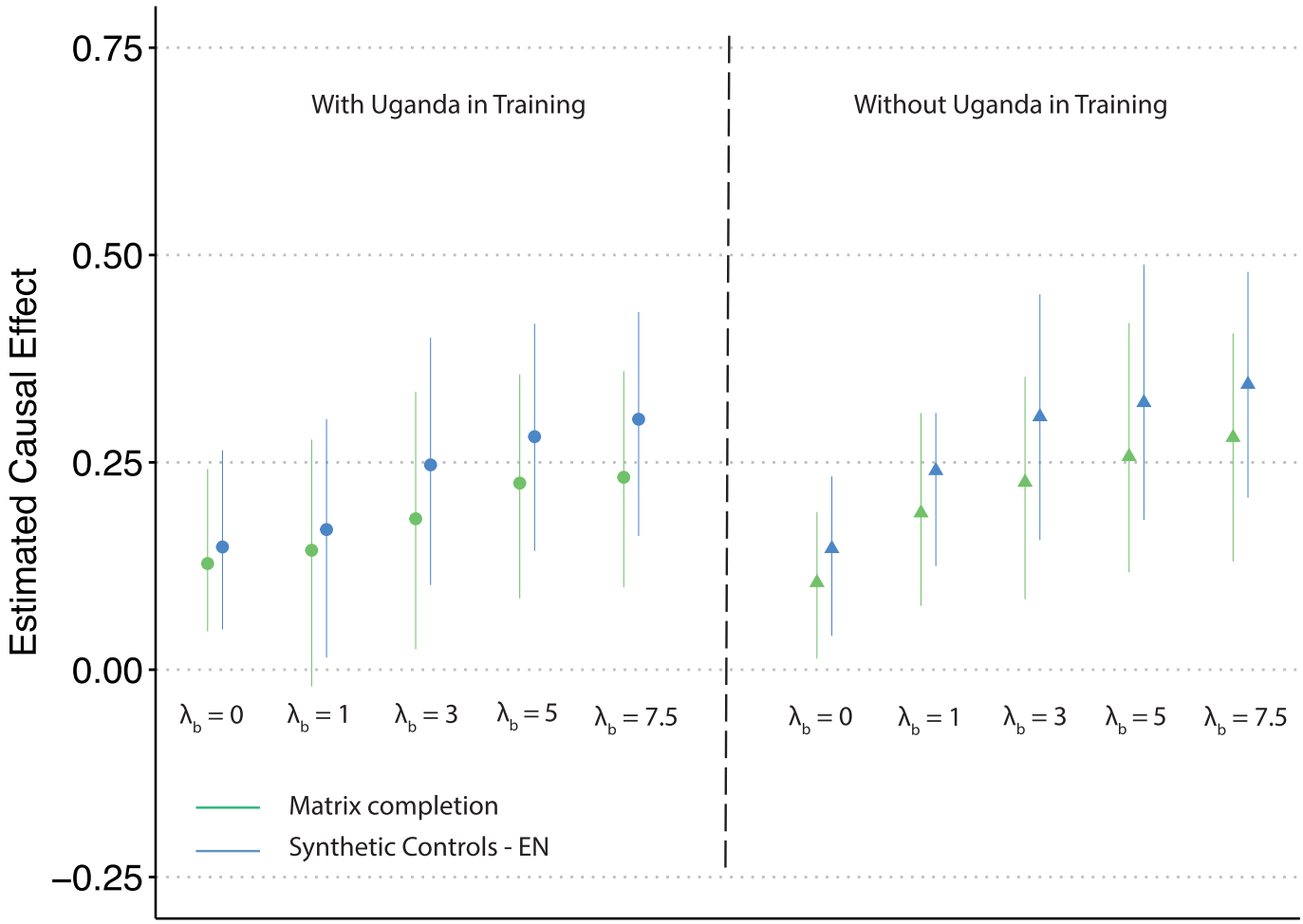
Extended Data Fig. 6 | Cross-validation shows that MC and SC-EN can accurately estimate the average of held-out control values. We split the control sample into ten equally sized, random folds and use MC, SC-EN and DiD to predict cluster-specific wealth values in the held-out test fold for each year, 2011–2016. We use each fold as a test set one time and in the training set nine times. The mean average differences across all years are 0.017, 0.018 and 0.013, respectively, signifying that MC and SC-EN can accurately predict unobserved values among control locations.



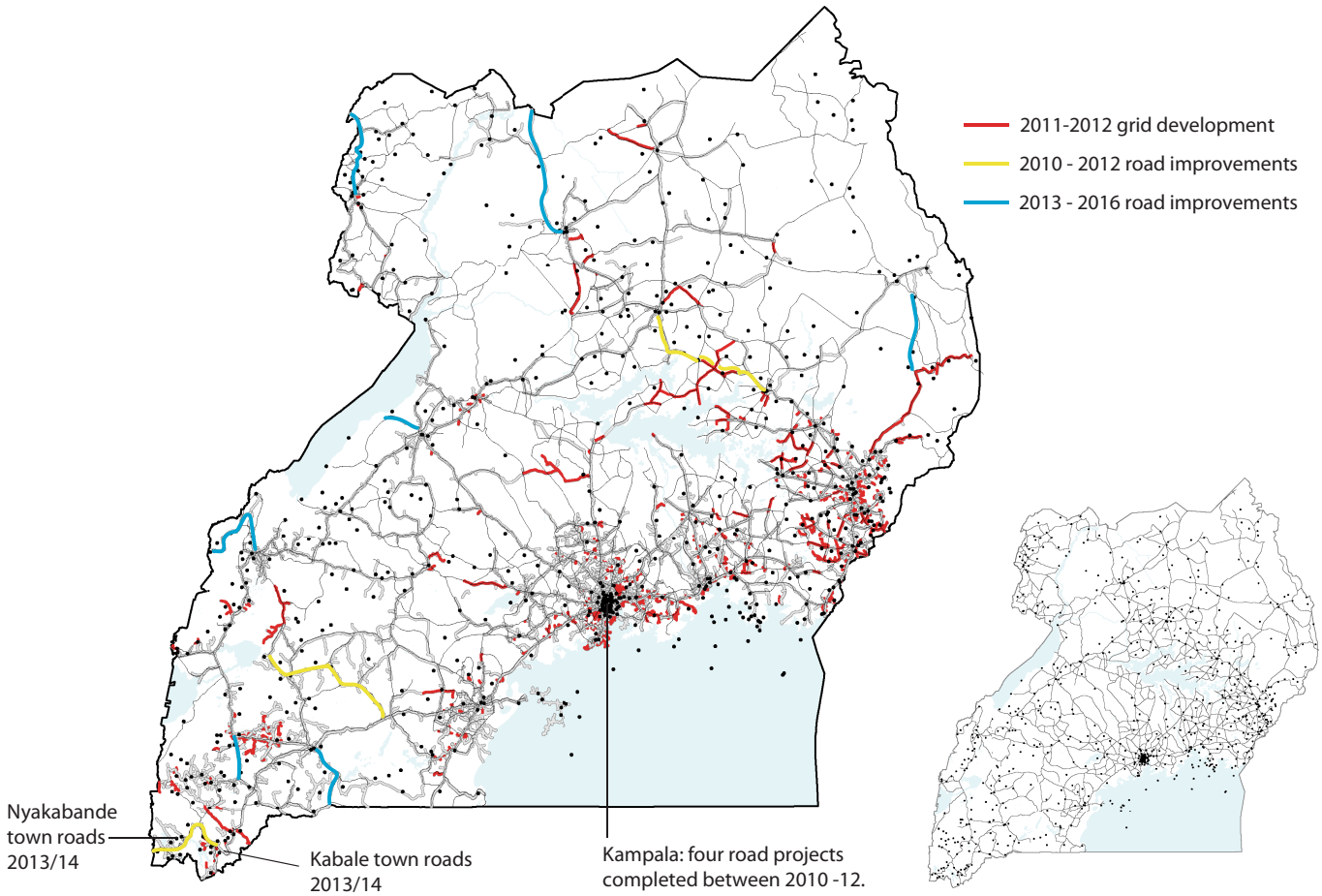
Extended Data Fig. 7 | Simulated data indicate that MC has smaller prediction error than DiD when pre-trend bias is present, and both estimators are equally attenuated under Berkson-type measurement error. a, We simulate a setting in which a treatment has a true-positive effect equal to 1 but units are trending differently in the absence of treatment and can have unit-specific trends that are correlated with treatment. We test this situation across several time periods (4 to 22 years, in which treatment begins in $t/2+1$) and varying growth rates of upward-trending units being selected into the treated group (0.1 to 0.25; darker blue or red lines indicate treated units with increasingly higher average growth rates). We then compare DiD and MC causal estimates and find that MC is less biased in the presence of treatment-correlated time-trending unobservables, particularly as time series lengthen. Blue dashed lines are DiD estimates that pool all years; blue solid lines are DiD estimates using only one post-treatment year; and red lines are MC estimates. **b**,**c**, We create a sample of 'observed' data that has a smaller variance than the true distribution, representing a common challenge in data imputation. Using our 'observed' sample data in **b**, we again simulate a setting that has a positive treatment effect equal to 1 but no differential pre-trends. Under this Berkson-type error, DiD and MC are similarly attenuated, underscoring the importance of bias reduction across the outcome distribution in the CNN training process.



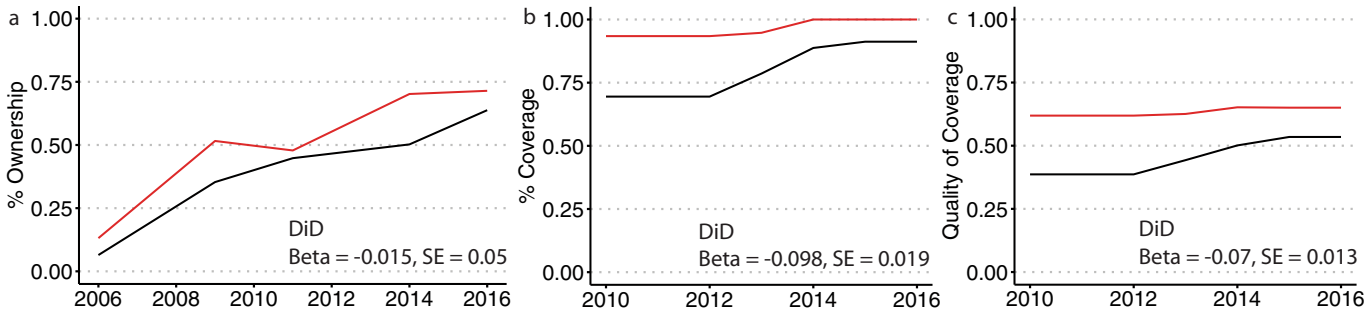
Extended Data Fig. 8 | Treatment effect estimates are robust to using larger inclusion buffers around electricity grid locations to identify treatment and control samples. Our baseline analysis uses a 2-km buffer around the electrical grid lines to identify treatment and control units. Treatment effect estimates using either a 3-km or a 4-km buffer are slightly larger but qualitatively similar, suggesting that our results are robust to plausible alternative buffers. Error bars represent 95% CIs, and are based on 100 bootstrapped model runs.



Extended Data Fig. 9 | The estimated treatment effect of electrification is higher when quintile-specific bias in wealth predictions is penalized more heavily. Each estimate represents a separate estimate from MC or SC-EN using output from a model with varying quintile-specific bias, from $\lambda_b = 0$ up to $\lambda_b = 7.5$, as in Extended Data Fig. 5. Error bars represent 95% CIs, and are based on 100 bootstrapped model runs.



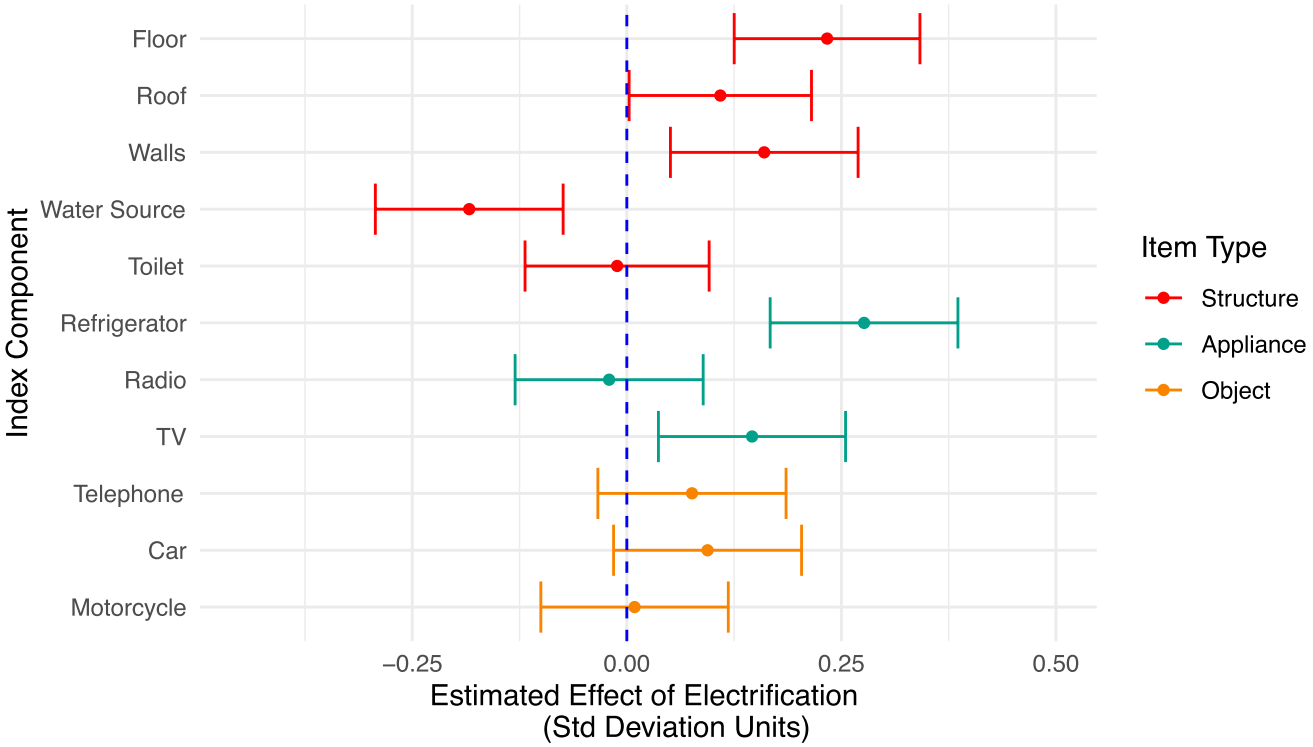
Extended Data Fig. 10 | Road improvements were modest in scale during our study period in Uganda and not associated with electrical grid extensions. Red lines denote electricity grid extensions made between 2011 and 2012, which are used to identify treated communities in our main analysis. Yellow lines denote road improvements (resurfacing) made before (2010) or during (2011–2012) our treatment period. Blue lines mark road improvements completed in the post-treatment period, and black dots in the main figure and inset panel represent DHS survey locations. During the entire period (2010–2016), road-upgrade projects averaged 184 km per year, totalling 1,102 km—less than 1% of the road network in Uganda. The location of road improvements shows almost no overlap with electricity grid development during our study period.



Extended Data Fig. 11 | Cellular phone ownership and coverage is not associated with electrical grid expansion. a, Cell phone ownership in newly electrified versus unelectrified communities based on household-level DHS responses in 2006, 2009, 2011, 2014 and 2016. A DiD estimate is shown in the lower-right corner, indicating that cell ownership in control communities grew faster post-treatment than in treated areas. **b**, Percentage of treated and control communities with cell phone coverage based on GSMA data from 2010 to 2016. Data again show that coverage grew faster in control communities post-treatment. **c**, As in **b**, except for the quality of cellular coverage ('variable' or 'strong' signal). Together, we do not find evidence of cellular ownership or investment being associated with new electricity access, suggesting that the public and private roll-out of cellular phone services did not occur in tandem with electricity grid development.

	Penalty	With UG, 95%	With Ug, 90%	Without UG, 95%	Without UG, 90%
1	0	6	7	10	14
2	1	5	13	8	11
3	3	4	12	19	32
4	5	8	13	22	29
5	7.5	3	8	15	25
6	5 (3km)	6	20	–	–
7	5 (4km)	9	18	–	–

Extended Data Fig. 12 | Rejection rate for DiD estimator in pre-trends test. Percentage of runs in which a DiD estimator rejected a null hypothesis of no difference in pre-treatment trends in asset wealth between treated and control locations, at either 95% or 90% confidence. 'With Uganda' and 'without Uganda' refer to whether or not Uganda was used in the training and validation steps. 'Penalty' indicates the quintile-bias penalty term (λ_b) used in each model run.



Extended Data Fig. 13 | Effect of electrification on specific assets in the asset index. Using DiD on DHS locations, we find statistically significant positive impacts across numerous individual assets, including three variables associated with home construction and three variables associated with electricity. We find weakly positive results for more expensive items, such as cars and motorcycles. Error bars represent 95% CIs.